



Emerald

△ Emerald stone featuring the signature emerald cut

One of the most desirable gemstones, emeralds are the rich green variety of beryl, the mineral found in igneous, metamorphic, and sedimentary rocks. Most emeralds have numerous inclusions and internal flaws, and these imperfections are unique to each stone. For jewellery, the brittle gem is usually faceted in its signature emerald cut. This is a step, or trap, cut, which combines a rectangular shape with shortened corner facets, maximizing the emerald's distinctive green colour, and protecting it from external damage and internal stress.

Specification

Chemical name	Aluminium beryllium silicate	Formula	$\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$
Colours	Emerald-green to green, yellow-green to blue		
Structure	Hexagonal	Hardness	7.5–8
SG	2.7–2.8	RI	1.565–1.602
Lustre	Vitreous	Streak	White
Locations	Colombia, Zambia, Brazil, Zimbabwe		

Rough



Columbia emerald | This finely formed, richly coloured, hexagonal emerald crystal is the size of a walnut. It originates from Santa Fe de Bogota, Colombia.

Cut



Extra face | The cutter of this octagonal, step-cut emerald has added an extra pavilion facet to aid the removal of a particularly bad internal flaw.



Emerald rough | The reddish staining on this specimen of emerald rough follows the lines of internal cracks, helping the cutter to assess its suitability.



Synthetic emerald | This synthetic pendeloque specimen has the same crystal structure as a natural emerald but can be purchased at a much lower price.



Emerald cut | Despite being relatively flawed, this octagonal emerald has been specifically shaped and cut in the signature style to minimize loss.

Emerald cut